

Intelligent Agent Lab

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In this hands-on lab, you'll put everything from this lesson into action by building your own intelligent agent using either LangChain or Rasa. Your agent should perform a multi-step task by integrating perception (input), decision-making (logic or planning), and action (tool or response). The task should involve at least two linked steps and one memory element or tool.

Objective

This activity brings together perception, planning, and action as you design a multi-step intelligent agent using LangChain or Rasa. You'll build real-world logic into your system—connecting memory, tools, and decision-making to complete a task. Is your agent just reactive—or truly reasoning? This is your first step toward architecting scalable, context-aware AI systems.

Tools Used:
LangChain or Rasa

Instructions for learners

Option A LangChain

- Build a simple research or utility agent
- Example: Given a user query, the agent fetches info using a search API, summarizes it, and stores the result in memory

Option B Rasa

- Build a dialog agent with structured conversation flow
- Example: A booking assistant that asks for the user's name and preferred time, confirms the appointment, and stores the data in a slot

Submission

What You'll Submit

- A brief description of your agent's purpose
- The steps your agent takes (including inputs, decisions, and outputs)
- Screenshots or snippets of your code or conversation flow
- A note on how memory or tool use supports the agent's decision-making

Consider pre-populating the prompt or expected toolstack (e.g., Colab + LangChain + OpenAI API).

Conclusion

Great work building your first intelligent agent! Now let's take a moment to reflect. In the next activity, you'll review your design decisions—why you chose certain tools, how your agent managed context, and where your logic could evolve. The Coursera Coach will help you evaluate whether your architecture is truly scalable and adaptable.